

Sample Questions - Regularization in DNNs and CNNs (AI2100/5100)

1. You are solving the binary classification task of classifying images as cat vs. non-cat. You design a CNN with a single output neuron. Let the output of this neuron be  $z$ . The final output of your network,  $\hat{y}$  is given as

$$\hat{y} = \sigma(\text{ReLU}(z))$$

where  $\sigma$  is sigmoid function. You classify all inputs with a final value  $\hat{y} \geq 0.5$  as cat images. What problem are you going to encounter? [2 Marks]

2. (a) Mention how the complexity of the mapping learned by the DNN increases with (i) depth of the model, and (ii) width of layers. [2 Marks]
3. Which of the following assertions are True about the convolution layer in a CNN? (identify multiple options if required) [2 marks]
- (i) Number of weights depend on the depth (or channels) of the input
  - (ii) Number of biases is equal to the number of filter in the layer
  - (iii) Number of parameters depends on the stride
  - (iv) Number of parameters depends on the padding
4. Consider a Neural network applied to industrial data with seven channels and  $16 \times 16$  spatial dimensions. Compute the number of parameters and products in the first layer for the following configurations. (You must consider biases also for the neurons, along with the weights. You may leave the expression as a product without finding the actual number, e.g.,  $5^2 \cdot 3^2 + 1$ )
- (a) Use a convolution layer with eight feature maps and filters of size 5 5 (use default values for stride and padding)
  - (b) Use a fully connected first layer where the number of input and output units are kept the same.
5. Which of the following is NOT a way to handle the vanishing gradient problem in CNNs when training them for classification?
- a) Using Auxillary classifiers in intermediate layers
  - b) Using Skip Connections, either by addition or concatenation
  - c) Using shallow architectures, but using more channels per layer
  - d) Using activations that do not zero out, e.g. LeakyReLU
6. What is the “receptive field” of a neuron (or activation) in a CNN performing a computer vision task?
7. Max Pooling with a stride of 2 does not increase the receptive field, because it discards values from the input. Instead, one should use a convolution with a stride of 2 to increase the receptive field. [True or False]
8. 1X1 convolutions in the inception block can be considered to perform an unsupervised dimensionality reduction. [True or False]
9. “Injecting a small amount of noise into the input to a neural network” is a trick that falls into the broad category of regularization known as —
10. Which of the following best describes the primary goal of regularization in deep learning?
- A. To reduce the generalization error by preventing the model from overshooting the noise in training data.

- B. To speed up the convergence of the gradient descent algorithm.
  - C. To decrease the training error as much as possible.
  - D. To increase the capacity of the neural network.
11. In L2 regularization (Weight Decay), what is the effect of the penalty term on the weights during backpropagation?
- A. It keeps weights small by subtracting a fraction of the weight from itself during every update.
  - B. It forces some weights to become exactly zero.
  - C. It ensures that the sum of the weights equals one.
  - D. It increases the weights of the most important features.
12. Why is Dropout only applied during training and not during the inference (testing) phase?
- A. The weights are frozen during testing, so dropout has no effect.
  - B. The test set is usually too small for dropout to be effective.
  - C. Applying it during testing would make the predictions stochastic and likely less accurate.
  - D. It is too computationally expensive to drop neurons during testing.
13. Which regularization technique is most useful when you suspect that only a few input features are actually relevant to the output?
- A. Data Augmentation
  - B. Dropout
  - C. L1 Regularization
  - D. L2 Regularization
14. How does Data Augmentation act as a regularizer?
- A. It reduces the number of parameters in the network.
  - B. It forces the model to learn invariant features that are robust to noise and transformations.
  - C. It simplifies the model architecture.
  - D. It automatically adjusts the learning rate.